CereBro – Functional Requirement Document (FRD)

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1. Document Purpose

The purpose of this FRD is to provide a **comprehensive functional blueprint** for CereBro, ensuring that development, design, and clinical teams align on **system behavior, user flows, and data handling** before development begins.

Key roles of this FRD:

1. **Clarity of Scope**
   - Clearly define what features the system must perform, how users will interact with it, and the conditions under which these features operate.
   - Prevent scope creep by distinguishing functional vs. non-functional requirements.
2. **Stakeholder Alignment**
   - Bridge between **business stakeholders (investors, founders, corporate clients)** and **technical implementers (engineering, AI, UX teams)**.
   - Provide a **common reference language** to ensure business intent is faithfully translated into system features.
3. **Compliance & Audit Preparedness**
   - Provide traceable documentation for regulators (CDSCO, GDPR, HIPAA) to verify that design decisions include user safety, consent, and data integrity.
   - Ensure clinical advisors can validate alignment with trauma-informed best practices.
4. **Development Readiness**
   - Define user journeys and system modules in enough detail for developers to begin building without ambiguity.
   - Outline **dependencies, integrations, and constraints** that must be factored into architecture.

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2. System Overview

CereBro is an **Adaptive Mental Wellness Continuum Engine**. It functions at the intersection of **psychology, AI, gamification, and enterprise wellness needs**, creating a platform that simultaneously addresses **individual well-being** and **corporate/insurance insights**.

2.1 System Layers

The platform is structured across **four interconnected layers**:

1. **Entry Layer**
    - **Onboarding**: ACE Questionnaire, Trauma Quest, baseline mood calibration.
    - **Fog Tunnel**: Initial "mystical" UX entry metaphor to reduce stigma and normalize first use.
    - **Mood Mirror**: Daily emotional check-ins disguised as gamified reflections.
2. **Narrative Interaction Layer**
    - **Journaling**: Guided expressive writing rituals with CBT + trauma-informed cues.
    - **Breath & Calm Loops**: HRV-driven regulation rituals with grounding animations.
    - **AI Companion**: Anima/Animus model providing archetypal guidance and tone-adaptive responses.
3. **Escalation Layer**
    - **Step 1**: AI Companion (daily rituals, light check-ins).
    - **Step 2**: AI Twin (therapist-trained model summarizing stress signals + interventions).
    - **Step 3**: Human Therapist Escalation (via integrated therapist marketplace).
    - All escalation framed narratively (e.g., "Oracle invites you deeper into the corridor").
4. **Enterprise Insight Layer**
    - **Corporate Dashboards**: Workforce stress heatmaps, attrition risk predictors, ESG reporting templates.
    - **Insurance APIs**: Risk scoring integration for underwriting models.
    - **Clinical Dashboard**: Therapist summaries + anonymized case briefs.

2.2 Platform Characteristics

- **Narrative-first UX**: Disguises clinical assessments as immersive story-driven rituals.
- **Agentic AI**: Distinct from generic chatbots by embedding Jungian archetypes and tone modulation.
- **Trauma-informed scaffolding**: ACE-based intake ensures system adapts to risk early.
- **Scalable SaaS core**: Supports both **B2C** user subscriptions and **B2B SaaS enterprise deployments**.
- **Global compliance-ready**: Designed to meet **GDPR, HIPAA, CDSCO** requirements simultaneously.

2.3 Target Environment

- **Platforms Supported**: iOS, Android, Web portal (enterprise dashboards).
- **Devices**: Smartphones (primary), wearables (HRV/GSR sensors), desktop (HR dashboards).
- **Hosting**: Cloud-native, multi-region (India, Singapore, EU, US).
- **Languages**: English (initial), Hindi & Mandarin (Phase 2), Spanish (Phase 3).

2.4 Constraints & Limitations

- **AI Role**: AI will not deliver medical advice; it only guides wellness rituals and nudges.
- **User Consent**: Explicit consent required for data sharing with insurers or corporates.
- **Offline Use**: Limited offline functionality (journaling, breathing rituals cached); AI requires connectivity.

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3. Functional Scope

The functional scope describes the **features and functions the system must provide** across B2C, B2B, and clinical dimensions.

3.1 Core User Modules (B2C)

3.1.1 Onboarding Module

- **Inputs**: User demographics, ACE Questionnaire (10 baseline Qs + parental ACE mapping).
- **Outputs**: Archetype assignment (Drifter, Thinker, Transformer, Seeker).
- **Functions**:
  - ACE scoring embedded covertly into onboarding.
  - Trauma Quest (guided story prompts for memory triggers).
  - Baseline emotional calibration via mood wheel.
- **Acceptance Criteria**: User profile must be initialized within <5 mins; archetype output stored securely.

3.1.2 Mood Mirror

- **Inputs**: Emoji reflection, journaling text, or voice recording.

- **Processing**:
  - Covert PHQ-9 / GAD-7 scoring embedded into flows.
  - Sentiment + hesitation analysis for risk detection.
- **Outputs**:
  - Immediate reflection ("Oracle reveals your shade today").
  - Longitudinal "mood scroll" (week/month view).
- **Acceptance Criteria**: Mood entry completed in <2 mins; covert scoring accuracy ≥85%.

3.1.3 Journaling Module

- **Inputs**: Text or voice-based journal entries.
- **Functions**:
  - Adaptive prompts (shadow work, grief work, attachment theory cues).
  - Sentiment scoring: positive, reflective, avoidance.
  - "Mirror Scroll" weekly summaries for user insight.
- **Acceptance Criteria**: Journaling must adapt prompt style within 2–3 entries based on tone trajectory.

3.1.4 Breath & Calm Loops

- **Inputs**: Breath pacing (on-screen animations), HRV sensor data (if wearable connected).
- **Functions**:
  - Box breathing, paced breathing, 4–7–8 breathing.
  - Visual rituals: expanding circles, "fog clearing."
  - HRV baseline tracking.
- **Outputs**: Stress regulation score + ritual completion badge.
- **Acceptance Criteria**: Ritual session runs uninterrupted offline; biometric data syncs once back online.

3.1.5 AI Companion (Anima/Animus)

- **Inputs**: User journaling + mood mirror data.
- **Functions**:
  - Conversational flow (text/voice).
  - Archetype-driven persona tone (Drifter → gentle guidance; Thinker → rational reframing).
  - Empathy mirroring ("I hear the shadow behind your words").
- **Outputs**: Daily ritual suggestions, narrative reflections, grounding nudges.
- **Acceptance Criteria**: Companion must adapt tone dynamically within 5 interactions.

3.1.6 Escalation Engine

- **Triggers**: Persistent high PHQ-9/GAD-7 scores, suicidal ideation keywords, HRV collapse.
- **Functions**:
  - AI Companion → AI Twin → Human Therapist path.
  - Escalation framed narratively, avoiding clinical alarm.
- **Outputs**: Therapist marketplace referral, with user consent.
- **Acceptance Criteria**: Escalation must trigger in <30s of risk detection; logs captured in compliance module.

3.2 Enterprise Modules (B2B)

3.2.1 Corporate Dashboard

- **Functions**:
  - Workforce mood trends (heatmaps).
  - Attrition risk predictor (AI model).
  - Journaling tone cluster analysis (anonymous).
  - ESG reporting auto-export.
- **Acceptance Criteria**: Dashboards refresh daily; no individual identifiers visible.

3.2.2 Insurance API

- **Functions**:
  - Provide risk scoring API (ritual adherence + stress marker averages).
  - HIPAA/GDPR compliant tokenized outputs.
- **Acceptance Criteria**: APIs must respond <500ms; no raw journaling text shared.

3.2.3 Clinical Dashboard

- **Functions**:
  - Escalation briefs: 1–2 page summaries of high-risk cases.
  - AI Twin caseload snapshots.
- **Acceptance Criteria**: Therapist access gated by explicit user consent.

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4. User Roles & Permissions

CereBro will serve multiple categories of users across B2C, B2B, and clinical contexts. Each role requires **specific access rights** and **permission boundaries** to maintain safety, privacy, and compliance.

4.1 User Role Matrix

Role	Primary Purpose	Access Rights	Restrictions
Individual User (B2C)	Daily mental wellness rituals & journaling	- Onboarding (ACE, Mood Mirror) - Journaling & Breath Loops - AI Companion conversations - Escalation consent	- Cannot see corporate dashboards - Cannot access anonymized aggregate data - Cannot override escalation triggers
Corporate HR / Employer (B2B)	Workforce insights & ESG reporting	- Anonymous dashboards - Stress heatmaps - ESG report exports - Attrition risk scores	- Cannot see individual-level data - No access to journaling text or mood logs - No user identifiers
Insurance Partner	Risk-based underwriting & wellness-linked incentives	- Aggregated wellness risk scores - Ritual adherence trends - API integrations	- No access to raw text/journals - Only anonymized, tokenized metrics
Therapist (Escalation Partner)	Handle escalated high-risk cases	- Case briefs generated by AI Twin - Consent-driven user profile (if user agrees)	- No access to full data lake - View limited to escalation context only
Admin (CereBro Ops)	Manage system configuration, compliance, and support	- Role management - Compliance/audit logs - Incident reports	- Cannot modify clinical/AI models without board approval - Restricted from viewing raw user journals

4.2 Permission Enforcement Rules

- **Least Privilege Principle**: Every role sees *only* what it needs.
- **Consent-First Model**: Users must explicitly opt-in before data can flow to HR/insurers/therapists.
- **Anonymization Gate**: Any corporate or insurer dashboard must run through anonymization APIs before rendering.

- **Audit Logs**: Every data access by HR, Insurer, or Therapist must be logged and reviewable.

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5. User Journeys

User journeys illustrate how different personas will interact with CereBro, across B2C and B2B contexts.

5.1 B2C User Journey – "Drifter" Archetype Example

1. **Onboarding**
    - User downloads CereBro.
    - Guided through "Fog Tunnel" entry metaphor.
    - Completes ACE Questionnaire + Trauma Quest.
    - Assigned Drifter archetype (playful, easily distracted).
2. **Daily Ritual Flow**
    - Opens Mood Mirror → selects emoji + voice reflection.
    - AI Companion responds with gentle nudge ("Your Oracle senses curiosity in your fog").
    - Offered short 3-minute breath loop ritual.
3. **Adaptive Journaling**
    - Prompt: "Write about the moment you felt unseen today."
    - AI detects avoidance tone → Companion gently reframes.
4. **Escalation Pathway**
    - After 7 days of persistent avoidance journaling, escalation is triggered.
    - Companion introduces "Oracle Corridor" → invites user to meet AI Twin.
    - If risk persists, therapist referral option unlocked.

5.2 B2C User Journey – "Thinker" Archetype Example

1. Onboarding assigns Thinker archetype (rational, analytical).
2. Mood Mirror prompts include logic-driven scales instead of emojis.
3. Journaling suggestions use structured CBT reframing (pros/cons lists).
4. Escalation framed as "Your Oracle suggests deeper reflection with a guide."

5.3 B2B User Journey – HR Leader

1. HR logs into CereBro dashboard.
2. Sees **stress heatmap** of departments (green/yellow/red zones).
3. Downloads **ESG compliance report** for Q2 submission.
4. Uses **attrition predictor** to flag at-risk teams.
5. Pushes anonymized report to leadership without exposing individual users.

5.4 B2B User Journey – Insurer

1. Insurance API pulls **wellness scores**:
    - % users completing 3+ rituals weekly.
    - Avg. journaling sentiment positivity.
    - Avg. HRV improvement trend.
2. Underwriting model adjusts premium discount.
3. No identifiable user data shared.

5.5 Clinical User Journey – Therapist (Escalation)

1. Receives **AI Twin-generated brief**:
    - Summarized journaling trends.
    - Mood Mirror progression.
    - Stress triggers & ACE profile snapshot.
2. Reviews escalation alert (e.g., suicidal ideation risk).
3. User gives **consent** for therapist engagement.
4. Therapist begins teleconsultation outside CereBro platform (partnered network).

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6. Functional Requirements

Each requirement is described with **ID, Description, Priority, Module, and Acceptance Criteria**.

6.1 Core Functional Requirements

ID	Requirement Description	Priority	Module	Acceptance Criteria
FR-01	User must complete ACE Questionnaire during onboarding	High	Onboarding	Must finish in <5 mins; archetype output correctly assigned

FR-02	Mood Mirror must covertly track PHQ-9/GAD-7	High	Mood Mirror	≥85% scoring accuracy against validated psych scales
FR-03	Journaling prompts must adapt dynamically	High	Journaling	Prompt tone changes within 2–3 entries based on trend
FR-04	AI Companion must adapt tone by archetype	High	AI Companion	Companion tone shifts across 4 archetypes
FR-05	Escalation must follow AI → AI Twin → Human flow	High	Escalation	Triggered <30s after risk detection; audit logged
FR-06	Corporate dashboards must be anonymous	High	B2B Dashboard	No user identifiers visible in HR view
FR-07	Insurance API must provide tokenized risk scores	High	Insurance API	API responds <500ms; HIPAA/GDPR compliant
FR-08	Clinical dashboard must summarize high-risk cases	Medium	Clinical Dashboard	AI Twin briefs generated automatically
FR-09	System must encrypt all journaling/mood entries	High	Security Layer	AES-256 encryption verified in audits
FR-10	Platform must handle 5K concurrent users	High	Infrastructure	Load-tested at 5K concurrent with <2s latency

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6.2 Additional UX/Functional Rules

- Every ritual (journaling, breath, companion chat) must end with a **grounding action** (e.g., breath reset).
- Escalation alerts must avoid clinical language ("Oracle calls you deeper" instead of "You are at risk").
- Corporate dashboards must refresh daily, not in real-time, to reduce misuse.
- Push notifications limited to **3 per day**, adapted to archetype tone.

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7. Assumptions & Dependencies

CereBro's development depends on external conditions and third-party systems. These must be documented to avoid bottlenecks later.

7.1 Key Assumptions

1. **Market & User Readiness**
   - Smartphones with minimum Android 11/iOS 15 will be the baseline.
   - Target users (18–45, urban, corporate) are comfortable with journaling, gamified apps, and guided meditation.
   - Corporate clients have budget allocations for wellness & ESG.
2. **Clinical Acceptance**
   - ACE Questionnaire, CBT reframing, expressive writing, and HRV tracking continue to be recognized as validated approaches.
   - Trauma-informed design is accepted by clinical advisors and regulatory bodies.
3. **Regulatory Climate**
   - India's CDSCO will allow categorization as a **wellness product**, not Class B medical device (initially).
   - GDPR/HIPAA frameworks will remain stable over 2–3 years.

7.2 Dependencies

1. **Technology**
   - Speech-to-text APIs for journaling (Google, Whisper).
   - Wearable integrations (Apple Health, Fitbit, Garmin, Oura).
   - Cloud services (AWS/GCP) with regional storage compliance.
2. **Regulatory & Clinical**
   - CDSCO, MAS (Singapore), and EU compliance approval.
   - Clinical partners for trial validation.
3. **Business**
   - Partnerships with HR/Insurance for pilots.
   - Seed funding ($500K) to support MVP build.
   - Availability of trained therapists for escalation network.

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8. Risks & Mitigations

8.1 Risk Matrix

Risk	Type	Impact	Likelihood	Mitigation Strategy
Unsafe AI outputs	Clinical/AI	High	Medium	Human-in-loop moderation; escalation thresholds; RLHF

(triggering, incorrect)				training on trauma-informed datasets.
User stigma → low adoption	Market	Medium	High	Narrative-first design (Oracle, archetypes); avoid medicalized UX.
Regulatory delays (e.g., CDSCO)	Compliance	High	Medium	Parallel rollout in Singapore; categorize as wellness initially.
Data breach / privacy failure	Security	High	Low	AES-256 encryption; tokenized APIs; biannual penetration tests.
Low retention after first week	Engagement	Medium	High	Daily nudges, gamified rituals, reward badges, archetype-driven prompts.
Employer misuse of dashboards	Ethics	High	Low	Strict anonymization; refresh dashboards only daily; audit logs.
AI latency or downtime	Technical	Medium	Medium	Offline journaling & breath loops; autoscaling backend clusters.
Funding shortfall	Business	High	Medium	Phased build (MVP → AI Twin → Enterprise); early pilots to generate traction.

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9. Success Metrics

CereBro's success will be tracked across **3 domains: user adoption, business growth, and clinical/ESG impact**.

9.1 User Adoption & Engagement

- **DAU/MAU ratio**: ≥30% within 12 months.
- **Weekly Ritual Completion**: 3+ per active user.
- **Day-30 Retention**: ≥25%.
- **Escalation Use**: ≥20% of users engage AI Twin or therapist.

9.2 Business Metrics

- **Year 1**: 10K MAUs, $300K revenue, 65% GM.
- **Year 2**: 60K MAUs, $1.2M revenue, 72% GM.
- **Year 3**: 150K MAUs, $4.5M revenue, 78% GM.

- **Enterprise Clients**: 3 by Phase 4 (HR/insurance).
- **B2C Conversion**: ≥20% of free tier → paid.

9.3 Clinical & ESG Impact

- **Validated Wellbeing Gains**: 15–20% average reduction in PHQ-9/GAD-7 scores during trials.
- **Corporate Productivity**: 10–15% reduction in stress-related absenteeism.
- **Insurance Uptake**: 1+ wellness-linked underwriting product by Year 2.
- **ESG Adoption**: ≥2 corporates integrate CereBro into ESG disclosures.

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10. Approval & Next Steps

10.1 Approval Workflow

- **Founders**: Ashok Bhattacharya, Ruchira Chakrabarty, Sandeep Bhattacharya, Raju Komaravolu.
- **Clinical Advisors**: Validate trauma-informed design & escalation safety.
- **Tech Leads**: Sign off scalability, AI feasibility, and integrations.
- **Investors**: Review BRD + FRD package for seed stage funding.

10.2 Next Steps (Execution Roadmap)

Phase 0: Preparation (Month 0–1)

- Freeze MVP features (Onboarding, Mood Mirror, Journaling, Breath Loops).
- Finalize Figma UX designs (96-screen flow).
- Confirm cloud infrastructure setup.

Phase 1: MVP Build (Month 2–5)

- Develop onboarding module + ACE integration.
- Deploy Mood Mirror + Journaling with covert PHQ-9/GAD-7 scoring.
- Build Breath Loops with offline fallback.
- Release closed beta for 500–1,000 users.

Phase 2: AI Companion & Twin (Month 6–9)

- Launch Anima/Animus AI Companion with archetype tone.

- Deploy AI Twin for escalation handling.
- Begin therapist network onboarding.

Phase 3: Regulatory & Trials (Month 10–12)

- Initiate CDSCO-compliant wellness categorization.
- Conduct small-scale corporate trial (1 HR partner).
- Begin insurance API pilot.

Phase 4: Enterprise Launch (Month 12–18)

- Full HR/ESG dashboards live.
- Secure 2–3 corporate clients.
- Expand to insurers.

Phase 5: B2C Scale (Month 18–24)

- Public launch with gamified virality.
- Expand multilingual support.
- Target 100K+ MAUs.